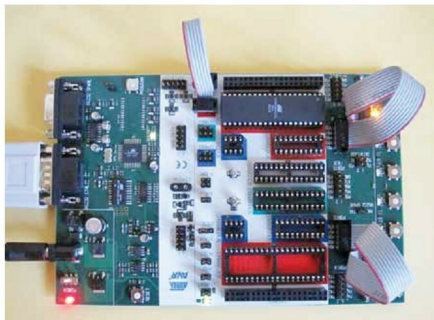


Which Development Board To Use For Your Embedded Project?

Given the plethora of development boards available in the market, selecting the one for your project may be a pretty mean task. However, once you see all these boards, there is a greater chance that you might find one that is more suited to your project at hand



ATMEGA STK500 board

DILIN ANAND

There are different choices of development boards available, whether you want to use an operating system (OS) or work on a pure embedded project. You would require a single-board computer (SBC) with enough random-access memory (RAM) and a fast central processing unit (CPU) to be able to run an OS like Linux. On the other hand, you could go for a microcontroller-based board with lots of interfaces if you are not using an OS.

Here is a look at some of the options you have.

Raspberry Pi

The Raspberry Pi is an ultra-cheap complete computer that can run Linux, Fedora, OpenELEC and Arch Linux. Developed by the Raspberry Pi Foundation to get the youth try their hand at engineering, it is a credit-card sized, bare-bones PC to which you can connect a TV, keyboard and mouse. On it, you can surf the Web, play Full-HD movies and do almost anything you could do on a normal PC—provided you have the patience to make it work.

The Raspberry Pi hardware consists of a 700MHz ARM11 CPU supported by a Broadcom VideoCore IV GPU. This SBC has a total of 256MB

SD RAM, which it generously shares with the GPU. The board is powered by a 5V microUSB or GPIO header and consumes between 500 mA (2.5W) and 700 mA (3.5W) depending on the model. The system-on-chip (SoC) that hosts the CPU, GPU, DSP and SD RAM is from Broadcom (BCM2835).

Another thing to note about the Raspberry Pi is that it does not come with a real-time clock. So an OS must use a network time server or ask the user for time information at boot time to get access to time and date information for file time and date stamping. However, a real-time clock (such as the DS1307) with battery backup can be easily added via the I²C interface. It is available from EFY associates Kits'n'Spares for ₹ 997. If you are buying in bulk, it could cost you as low as ₹ 615.

Arduino

Arduino is one of the most popular microcontroller-based boards used in robotics. Just like there are two models of the Raspberry Pi, Arduino has a large number of models that are differentiated based on the size, power consumption and processing capabilities. However, all Arduino boards use the ATMEGA AVR series microcontrollers from ATMEGA, which feature both analogue and digital pins. Another common feature is the software. An Open Source, 'C'-based software, called Arduino, can be used to program any Arduino board. Users are free to use any other software to program the microcontrollers.